

Tejaswini A
Senior Data Engineer
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Professional Summary

- Data Engineer with **5+ years of experience** designing and supporting **scalable, secure, high-performance data platforms** across **Healthcare** and **Banking** domains.
- Strong expertise in end-to-end **ETL/ELT** pipelines, cloud data lakes, and data warehouses using **Azure Data Factory, Azure Databricks, Apache Spark, Snowflake, Microsoft Fabric, and AWS**.
- Experience enabling **AI** and **GenAI**-ready data platforms by building feature pipelines, **RAG**-ready datasets, and governed data foundations for **ML** and **LLM**-based analytics.
- Proven experience delivering **HIPAA**-compliant healthcare analytics for claims and member data, supporting audit readiness and regulatory reporting.
- Hands-on experience supporting **banking and financial data systems**, enabling accurate transaction, loan, and risk analytics.
- Extensive background in retail analytics, processing sales, inventory, pricing, and clickstream data using **batch** and **near real-time pipelines**.
- Strong analytics and reporting expertise using **SQL Server, SSRS, Power BI, and Tableau** to deliver operational and executive dashboards.
- Proficient in automation and **CI/CD** driven data engineering using **Python, Terraform, Docker, Kubernetes, Jenkins, and Azure DevOps**.
- Effective collaborator in **Agile/Scrum environments**, partnering with **business, compliance, and analytics teams** to deliver **production-grade data solutions**.

Technical Skills

Data Engineering & Big Data: Apache Spark, Spark SQL, PySpark, Apache Kafka, Spark Structured Streaming, Delta Lake, Lakehouse Architecture, Medallion Architecture, Hive (SQL)

Distributed Data Platforms: HDFS, YARN

Cloud Data Platforms & Warehouses: Snowflake, Azure SQL Database, SQL Server, Amazon Redshift, Synapse Analytics, Oracle, Teradata, DB2, PostgreSQL

NoSQL & Specialized Datastores: MongoDB, Cassandra, HBase

Data Integration & Orchestration: Azure Data Factory, Microsoft Fabric Data Factory, AWS Glue, Apache Airflow, Fivetran, Matillion, Boomi, DBT

Cloud Technologies

Microsoft Azure: Azure Databricks, Azure Data Lake Gen2, Synapse Analytics, Microsoft Fabric (Lakehouse, One Lake), Azure Key Vault.

Amazon Web Services (AWS): S3, EMR, Glue, Redshift, Athena, Lambda, EC2, CloudFormation, IAM

Programming & Query Languages: Python, SQL, PySpark, T-SQL, PL/SQL

Streaming & Real-Time Processing: Apache Kafka, Spark Structured Streaming, Event-Driven Architectures, CDC concepts

Analytics, BI & Semantic Layer: Power BI (DAX, semantic models), Tableau, SQL-based analytics

DevOps, CI/CD & Platform Engineering: Terraform (IaC), Docker, Kubernetes, Jenkins, Azure DevOps (ADO), Git-based workflows, CI/CD for data pipelines

AI-Ready Data Engineering & GenAI Enablement: RAG-ready data pipelines, embedding-ready datasets, feature engineering for ML/GenAI, ML flow

Operating Systems: Linux, UNIX, Windows

Experience

Client: Optum, TX
Data Engineer

May 2024 to Present

Responsibilities:

- Owned end-to-end design and production support of a Lakehouse architecture using **Azure Databricks**, **Azure Data Lake Gen2**, and **Delta Lake** for healthcare claims, enrollment, eligibility, and provider data.
- Built Bronze, Silver, and Gold (Medallion Architecture) layers, enabling scalable ingestion, transformation, and analytics-ready datasets using **PySpark** and **Spark SQL**.
- Developed incremental and CDC-style data pipelines, handling late-arriving data and schema evolution to maintain accurate and consistent healthcare reporting.
- Delivered curated datasets to **Snowflake**, **Azure SQL**, and **SQL Server**, supporting **Power BI** semantic models (DAX) and **SSRS** regulatory and operational reports.
- Designed fact and dimension data models for claims, members, and providers to enable standardized analytics and reporting across enterprise healthcare use cases.
- Built AI-ready data pipelines on **Azure Databricks** and **Delta Lake** to prepare structured and semi-structured healthcare data for downstream **ML** and **GenAI** use cases.
- Developed feature engineering workflows using **PySpark** and **Spark SQL** to generate clean, standardized, analytics-grade datasets consumed by data science and AI teams.
- Created **RAG**-ready datasets by performing data chunking, normalization, metadata enrichment, and deduplication to support **LLM**-based search and summarization use cases.
- Prepared embedding-ready datasets by ensuring data quality, schema consistency, and versioned storage across **Delta Lake** and **Snowflake** for vector-based retrieval systems.
- Implemented data quality and reconciliation frameworks using **Spark SQL** and **SQL** to validate claims and eligibility data, ensuring audit readiness and regulatory accuracy.
- Optimized **Spark** and **Databricks** workloads through partitioning strategies, Delta Lake file compaction, and query tuning to improve pipeline performance and reduce processing time.
- Enforced **HIPAA** compliance through data masking, encryption, RBAC, and secure access controls across data engineering, analytics, and **AI-consumed datasets**.
- Orchestrated and monitored pipelines using **Azure Data Factory** and **Databricks** workflows, improving reliability, observability, and operational efficiency.
- Collaborated closely with business analysts, healthcare SMEs, compliance, and data science teams to translate regulatory, analytics, and AI data requirements into scalable data engineering solutions within Agile/Scrum environments.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Gen2, Delta Lake, Snowflake, Azure SQL Database, SQL Server, Apache Spark, PySpark, Spark SQL, Microsoft Fabric (Lakehouse), Python, Power BI (DAX), SSRS, Feature Engineering Pipelines, RAG-ready Data Pipelines, Embedding-ready Datasets, Healthcare Claims & Member Data, HIPAA Compliance, Data Quality & Reconciliation, ETL/ELT, Agile/Scrum.

Client: PNC, TX
Data Engineer

Sep 2023 to Apr 2024

Responsibilities:

- Served as the **primary data engineer** for banking analytics pipelines, owning ingestion, transformation, validation, and deployment workflows.
- Designed and supported **cloud-based** banking data pipelines for transaction, customer, and loan datasets using **Azure Data Factory**, **Azure Databricks**, and **Snowflake**.
- Integrated data from multiple banking source systems (core banking, loan servicing, customer systems) into centralized cloud data platforms, ensuring consistent data definitions and improved reporting accuracy.
- Developed Spark-based (**PySpark**, **Spark SQL**) transformations to process batch and incremental financial data, applying **CDC concepts** to ensure accurate and timely data availability.
- Implemented incremental and historical data processing patterns to support point-in-time reporting for loans, balances, and customer analytics.

- Built and maintained data quality validation and reconciliation checks to ensure accuracy for **regulatory, compliance, and risk reporting** requirements.
- Designed analytics-ready fact and dimension data models in **Synapse Analytics and Snowflake**, enabling consistent **Power BI** semantic models and **SQL**-based analytics.
- Optimized **Spark** and **Snowflake** workloads through query tuning and transformation improvements to enhance pipeline performance and reliability.
- Supported secure data access and governance practices by enforcing role-based access controls and adhering to internal banking compliance and audit standards.
- Supported **CI/CD** workflows and source control using **Azure DevOps**, ensuring controlled deployments, versioning, and environment consistency.
- Collaborated with **business analysts, finance teams, and risk stakeholders** to translate banking and regulatory requirements into scalable data engineering solutions.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Gen2, Snowflake, Apache Spark, PySpark, Hive, HDFS, Python, SQL (MS-SQL), Microsoft Fabric, Tableau, Data Warehousing, ETL/ELT, Agile

Client: Cognizant
Data Engineer

Oct 2019 - July 2023

Responsibilities:

- Designed and implemented cloud-native data platforms on **AWS** using **S3, EMR, Glue, Redshift, and Athena**, supporting large-scale batch and near real-time analytics workloads.
- Built and optimized cloud data warehouses on **Amazon Redshift** and **Snowflake**, developing **Python, SQL, and Spark-based ETL/ELT pipelines** for efficient data ingestion and transformation.
- Developed **Apache Spark** pipelines to process millions of transactional and clickstream records daily, supporting near real-time retail analytics and reporting.
- Implemented event-driven and near real-time data pipelines using **Apache Kafka** and **Spark** Structured Streaming, writing curated outputs to **Snowflake** and **Cassandra**.
- Orchestrated and monitored data workflows using **Apache Airflow**, ensuring reliable scheduling, dependency management, and operational visibility.
- Managed metadata and schema governance using **AWS Glue Data Catalog**, enabling consistent schema discovery and evolution across the data lake.
- Performed **data migration** and **modernization** from legacy platforms (**Teradata, DB2, Oracle**) to cloud data lakes and warehouses using **Hive, HDFS, and S3**.
- Tuned **Spark** and distributed processing workloads by optimizing partitioning, memory usage, and query execution to improve pipeline performance and stability.
- Supported **CI/CD**-enabled data platforms using **Terraform (IaC), Docker, Kubernetes, Jenkins, and Git**-based workflows to enable repeatable and automated deployments.
- Built **SQL**-based analytics and **Tableau** dashboards on curated datasets to support business reporting and operational insights.

Environment: AWS (S3, EMR, Glue, Redshift), Apache Spark, PySpark, Snowflake, Kafka, Apache Airflow, Hive, HDFS, SQL, Data Warehousing, ETL/ELT, CI/CD, Agile

Education

Masters in University of North Texas